

## **FSD-2 PROJECT**

### **Skill Exchange Platform for Students**

# **PeerLink**

## **A Peer-to-Peer Learning and Skill Sharing Platform for College Students**

**Domain: Education Technology**

**Stack: MERN (MongoDB, Express.js, React.js, Node.js)**

### **1. Problem Statement**

- Students possess various technical and non-technical skills but lack a centralized platform to exchange knowledge.
- Finding peers who are willing to teach or learn a particular skill is difficult.
- Existing learning platforms are mostly paid and do not encourage peer-to-peer skill sharing.
- Students are unable to showcase their skills effectively within their college community.
- There is a need for a platform that promotes collaborative learning and skill development.

### **2. Team Details**

1. Shaik Naaziya – 24B01A45B3 (Team Lead)
2. Shaik Fazrin – 24B01A45B2
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### **3. Abstract**

The Skill Exchange Platform for Students is a web-based application designed to promote peer-to-peer learning among college students. The platform enables students to showcase their skills, connect with peers who are interested in learning them, and exchange knowledge in a collaborative environment. Students can create skill profiles, search for other users based on their interests and expertise, and send skill exchange requests. By encouraging skill sharing within the student community, the platform aims to enhance practical learning, networking opportunities, and continuous self-

development. Built using the MERN stack, the project provides a secure, scalable, and user-friendly solution for collaborative learning.

#### 4. Objectives

- Develop a web-based skill exchange platform for students.
- Enable students to create skill profiles.
- Allow students to teach and learn skills from peers.
- Facilitate skill exchange requests and approvals.
- Promote collaborative and practical learning.
- Build an active student learning community.

#### 5. Approach / Modules

##### **Student Module**

- Register and log in.
- Create and update skill profiles.
- Add skills they can teach.
- Add skills they want to learn.
- Search other students based on skills.
- Send skill exchange requests.
- View accepted and pending requests.

##### **Skill Exchange Module**

- Match students based on skills.
- Manage exchange requests.
- Schedule learning sessions (optional).

##### **Admin Module**

- Manage users.
- Remove inappropriate content.
- Monitor platform activities.
- View reports and statistics.

#### 5. Functional Requirements

The system shall:

- Allow students to register and log in securely.
- Allow users to create and manage their skill profiles.
- Enable students to add skills they can teach and skills they wish to learn.
- Allow users to search for students based on specific skills.
- Enable students to send and manage skill exchange requests.
- Allow administrators to monitor and manage platform activities.
- Maintain user authentication and data security.

## 6. Technologies Used

| Component         | Technology             |
|-------------------|------------------------|
| Frontend          | React.js, Tailwind CSS |
| Backend           | Node.js, Express.js    |
| Database          | MongoDB, Mongoose      |
| Authentication    | JWT, bcrypt.js         |
| API Testing       | Postman                |
| Version Control   | Git & GitHub           |
| Development Tools | VS Code                |

## 6. Purpose of Technologies Used

### React.js

- Builds responsive user interfaces.
- Provides component-based development.

### Express.js

- Creates REST APIs.
- Handles backend operations.

### MongoDB

- Stores user and skill data.

### JWT

- Provides secure authentication.

### bcrypt.js

- Encrypts passwords before storing them.

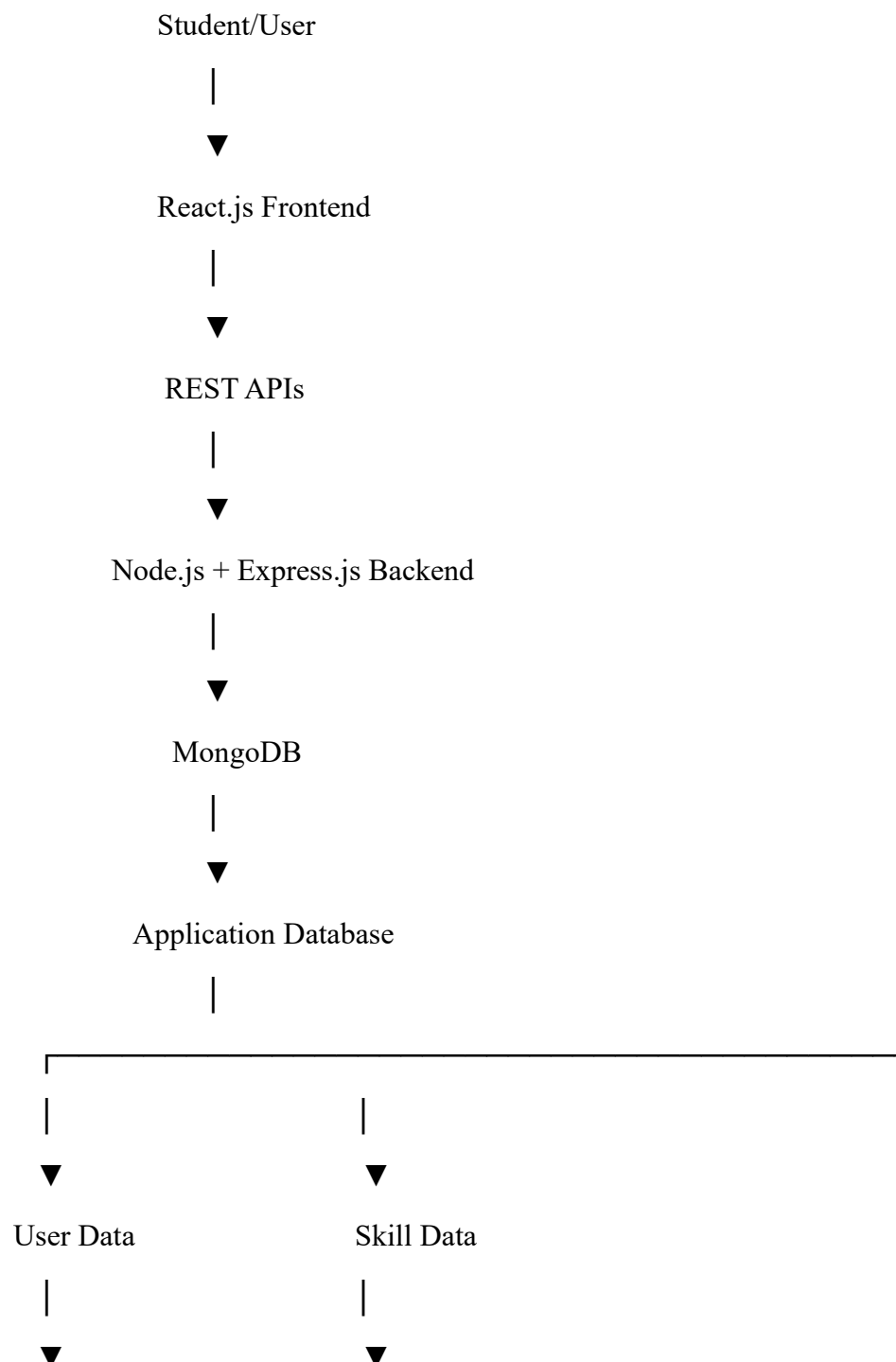
### **Tailwind CSS**

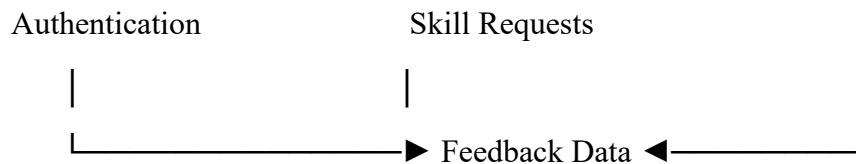
- Makes UI responsive and attractive.

### **Git & GitHub**

- Used for version control and collaboration.

## **7. System Architecture**





## 8. database Collections

### Users Collection

- User ID
- Name
- Email
- Password (Encrypted)
- Skills Offered
- Skills Required
- Profile Information

### Skill Requests Collection

- Request ID
- Sender ID
- Receiver ID
- Skill Name
- Request Status

### Feedback Collection

- Feedback ID
- User ID
- Rating
- Comments

## 9. Expected Outcome

- Students can easily share and acquire new skills.
- Promotes peer learning within colleges.
- Helps students build practical knowledge.

- Creates networking opportunities among students.
- Improves skill development and collaboration.
- Encourages continuous learning.

#### **8. Future Enhancements**

- Video call integration.
- AI-based skill recommendations.
- Certificate generation after skill exchange.
- Integration with college ERP systems.
- Chat and notification system.
- Skill assessment quizzes.